IEEE BIP 2026 · GUÁPILES, COSTA RICA · 11–13 NOVEMBER

BeeQ: Evaluating Quantum Fidelity Kernels as a New Approach Methodology for Molecular Toxicity Prediction

MATCHED COMPARISON | 893 MOLECULES | 10 SIMULATED QUBITS

Luis A. Zumbado-Silva
CBio3 Laboratory, School of Chemistry, UCR

Johnny Z. Agüero-Sandí
School of Computer Engineering, TEC

Mauricio Gutiérrez-Arguedas
School of Chemistry, UCR

Andrés Quesada-González
School of Computer Science and Informatics, UCR

Melissa Carvajal-Charpentier
School of Computer Science, TEC

William J. Zamora-Ramírez
CBio3 Laboratory, School of Chemistry, UCR

AGENDA

01 **Context**

Why honey-bee toxicity matters, and the research question that drives the work.

02 **Methodology**

The dataset, how molecular families are separated, and how each molecule is represented.

03 **Implementation**

What a kernel is, how descriptors are encoded into quantum states, and the scope of the simulation.

04 **Results**

Performance across model families, transfer beyond Development, and the external CR8 panel.

05 **Conclusions**

What the study does and does not show, and where it goes next.

There are far more substances than feasible assays

01 · EXPOSURE

Apis mellifera is a key pollinator

Agricultural active ingredients can reach it, and the acute lethal dose is an established regulatory criterion.

02 · MEASUREMENT

Measuring it requires animal testing

The in vivo assay gives the LD_{50} per bee, but it is slow, costly, and does not scale to the number of registered substances.

03 · PRIORITISATION

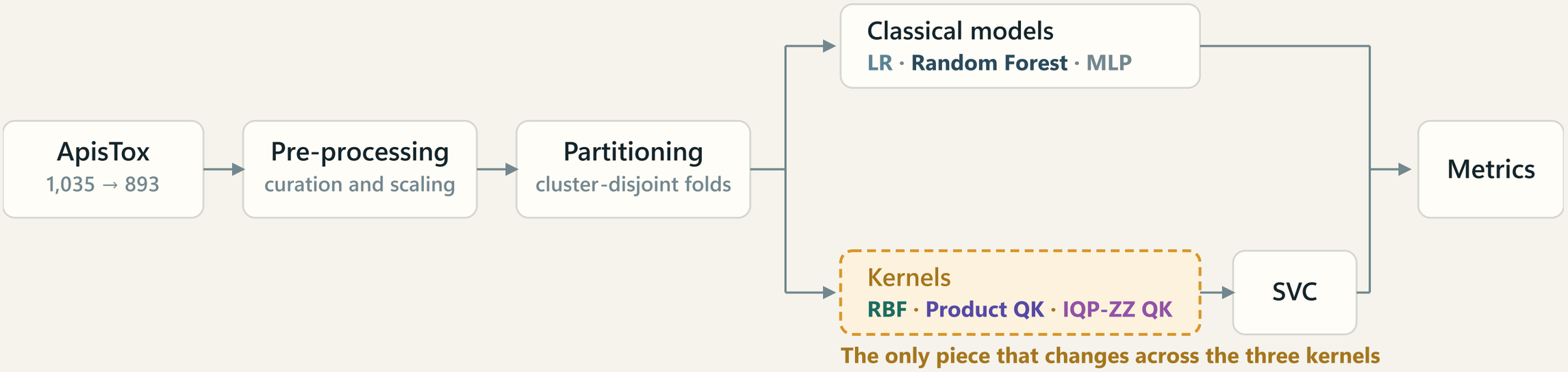
Not every molecule: the ones that matter

Models trained on already measured data flag which molecules are worth taking to the laboratory and which are not.

BeeQ sits at the third step: an exploratory prioritisation tool, not a regulatory decision criterion.

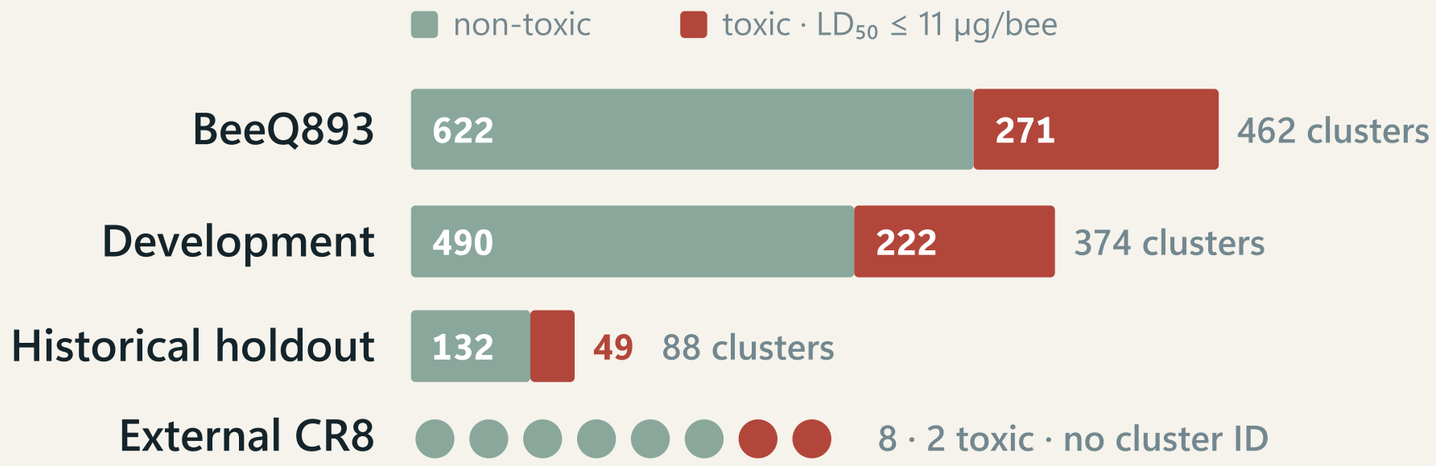


Does a fixed quantum fidelity kernel induce a predictively and geometrically different similarity structure when the molecular representation, partitions, pre-processing and evaluation are kept unchanged?



--- Conceptual schematic

893 curated molecules with measured acute toxicity



SI Table S1 · frozen campaign

- **From 1,035 to 893**

  ApisTox contributes 1,035 compounds; additional curation and standardisation with RDKit leaves 893 organic molecules.

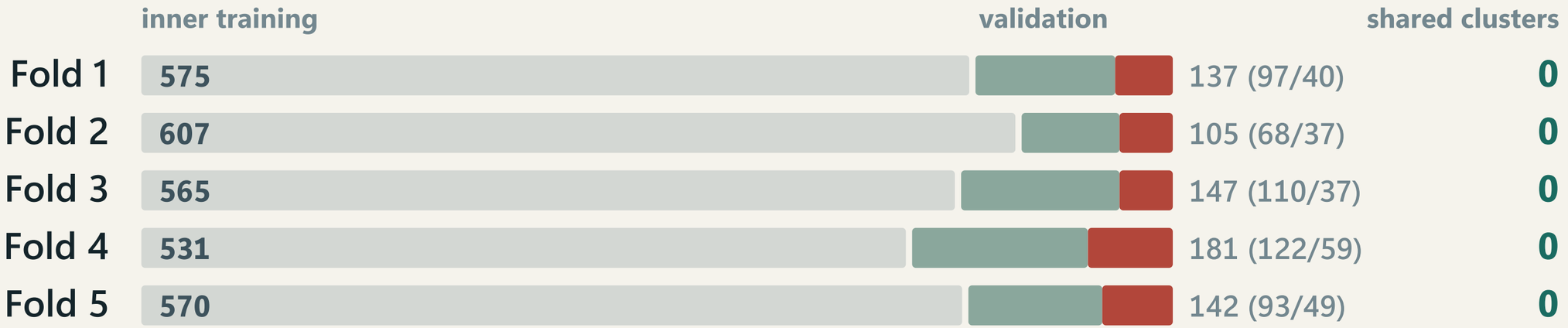
- **Binary classification, not regression**

  The threshold was fixed before modelling and never tuned on performance.

THE ENDPOINT

$$y = 1 \text{ if } LD_{50} \leq 11 \text{ µg/bee}$$
$$y = 0 \text{ if } LD_{50} > 11 \text{ µg/bee}$$

Molecular families are never split across sides



WITHIN EACH OUTER FOLD

i1 i2 i3 i4

4 cluster-grouped splits

→

Selection

hyperparameters by AUROC
threshold by maximum MCC

→

FROZEN

one refit on the 712
+ final threshold

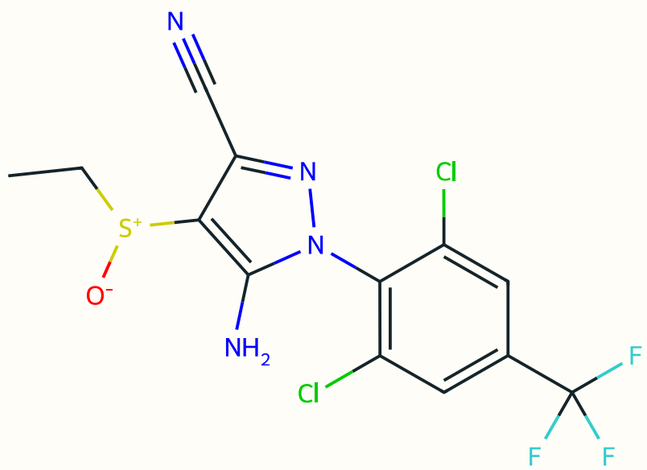
→

Holdout · 181

CR8 · 8

evaluated once

From a chemical structure to ten numbers



Ethiprole

— Structure drawn with RDKit from the frozen CR8 SMILES

logP 1 **3.78** z +0.44 θ +0.109	**MW** 2 **397.21** z +0.77 θ +0.193	**TPSA** 3 **90.69** z +0.37 θ +0.092	**HBD** 4 **1** z +0.12 θ +0.031
RotB 5 **3** z −0.52 θ −0.130	**ArR** 6 **2** z +0.78 θ +0.195	**nHal** 7 **5** z +1.94 θ +0.486	**nOP** 8 **0** z −0.36 θ −0.090
LiPHEX 9 **−1.92** z −1.62 θ −0.406	**fSASA** 10 **0.19** z +0.27 θ +0.067		

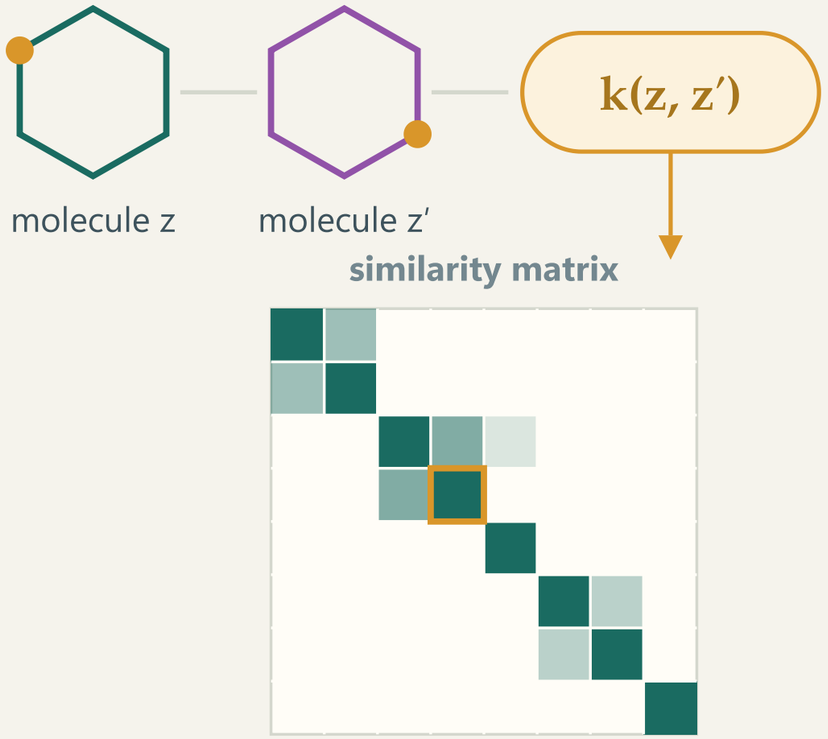
- **value** of the descriptor computed on the structure
- **z** **standardised:** (value − mean) / SD, with Development mean and SD
- **θ** **angle:** θ = s · z, what rotates the qubit, with s = 0.25

A kernel is a way of measuring likeness

Instead of asking the model to reason about each molecule on its own, we hand it a table with the likeness of **every pair** of molecules.

$$K_{RBF}(z, z') = \exp(-\gamma \, \|z - z'\|^2)$$

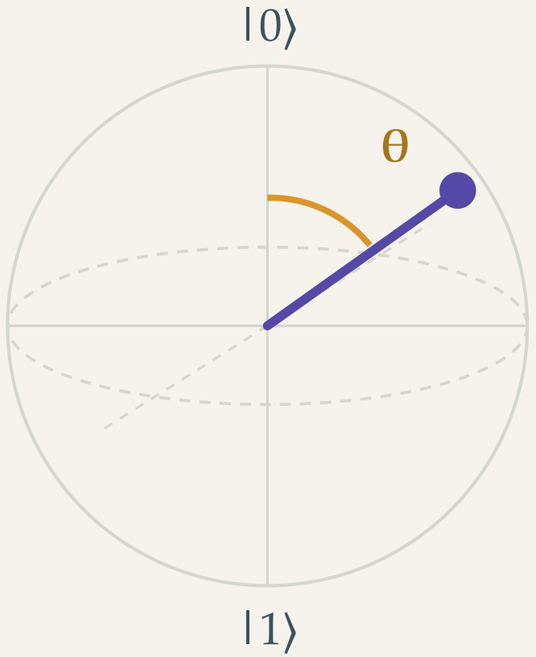
The classical reference. It decays with the distance between standardised descriptors; γ controls how fast.



One pair of molecules fills one cell.

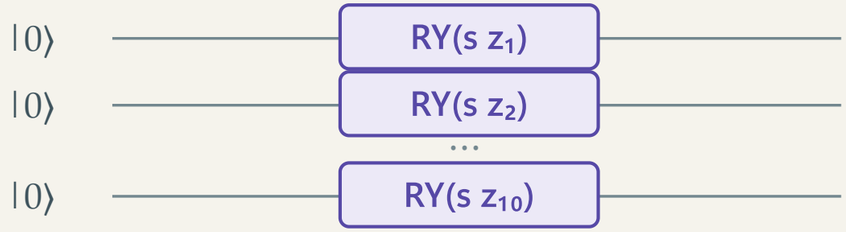
Each descriptor rotates a qubit; likeness is the fidelity

LIKENESS = FIDELITY $K_Q(z, z') = |\langle\psi(z)|\psi(z')\rangle|^2$ Between 0 and 1; it is 1 when the two states coincide.



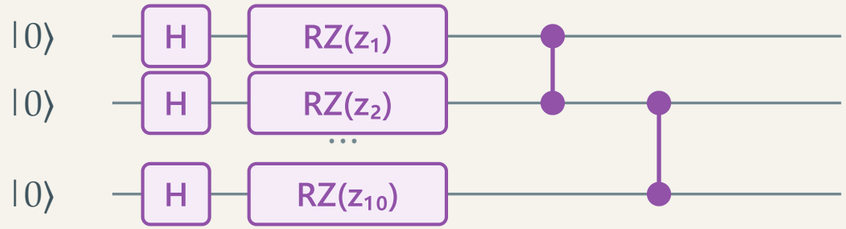
$\theta_i = s \cdot z_i$ · one qubit per descriptor

PRODUCT · s = 0.25



no entanglement

IQP-ZZ · s = 0.125 · λ = 1

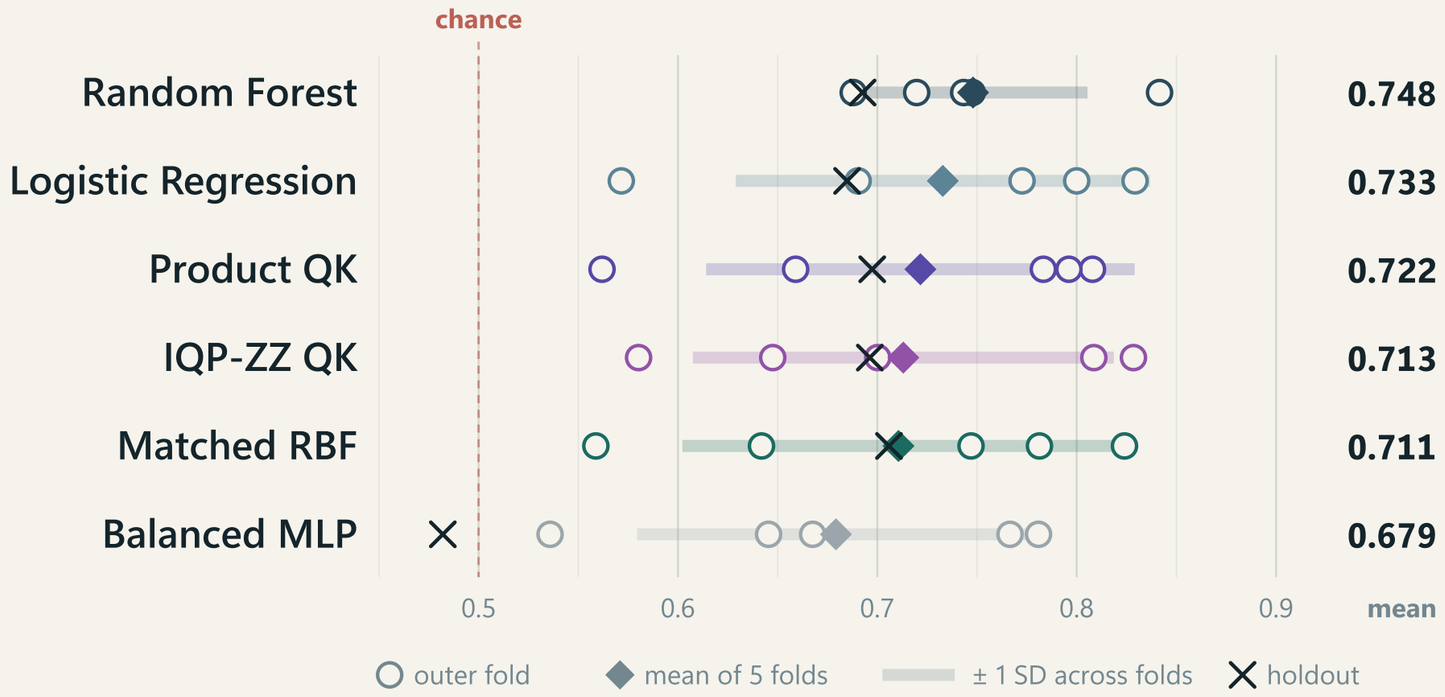


RZZ chain
nearest neighbours
no final Hadamard

--- Conceptual schematics

- Exact statevectors in NumPy
- 1,024 amplitudes per molecule
- No shots, no noise, no quantum hardware

Dispersion across folds exceeds the differences between models



Random Forest reached the highest mean: 0.748

AUROC: ranking, not accuracy

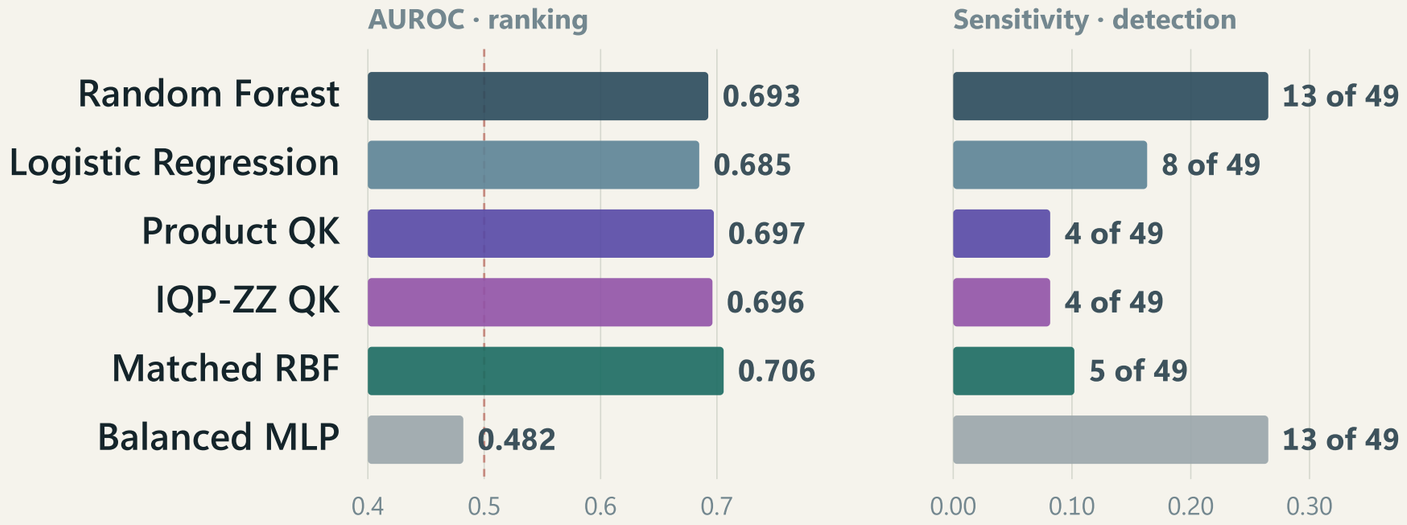
Probability that a random toxic molecule scores above a random non-toxic one.

The three kernels landed together

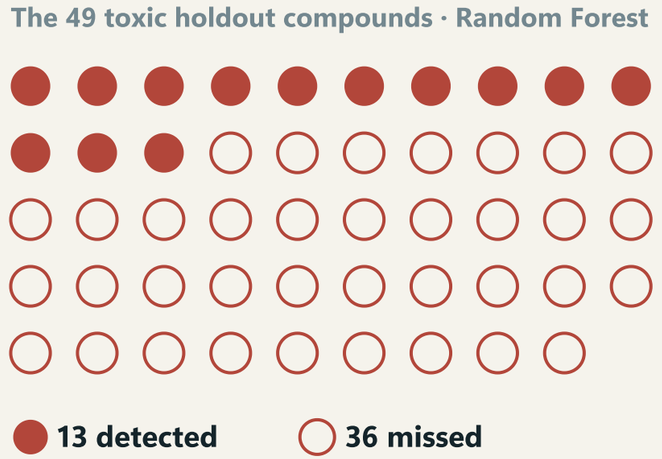
Product 0.722 · IQP-ZZ 0.713 · RBF 0.711, with fold-wise SDs of 0.106 to 0.108.

— Paper Table I · SI Tables S17 and S18

Ranking well and detecting little are compatible

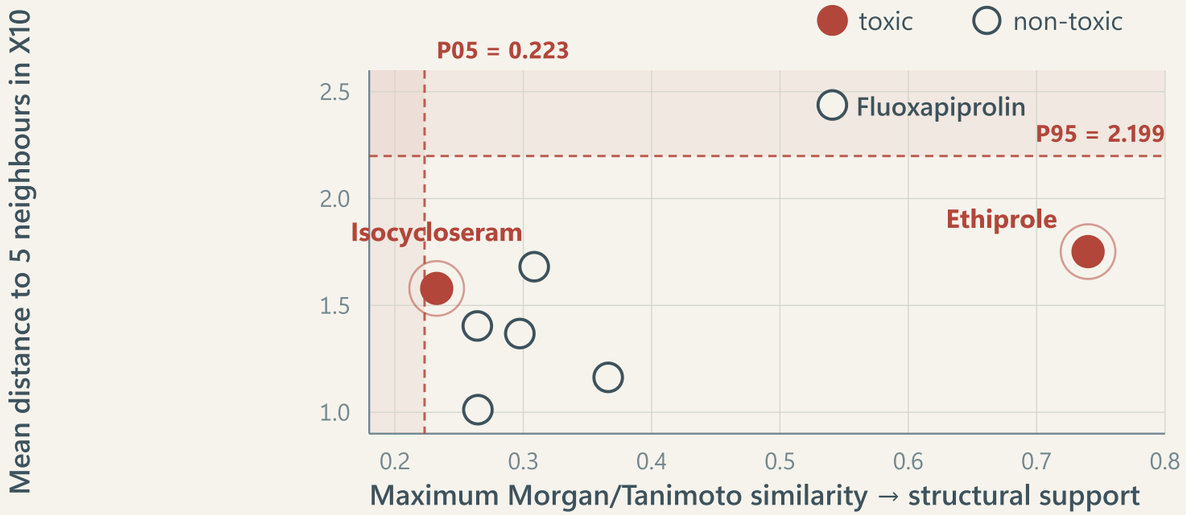


— SI Table S18



Five models hold AUROC near 0.70 and still let most of the toxic compounds through.

Being inside the domain does not guarantee a correct call



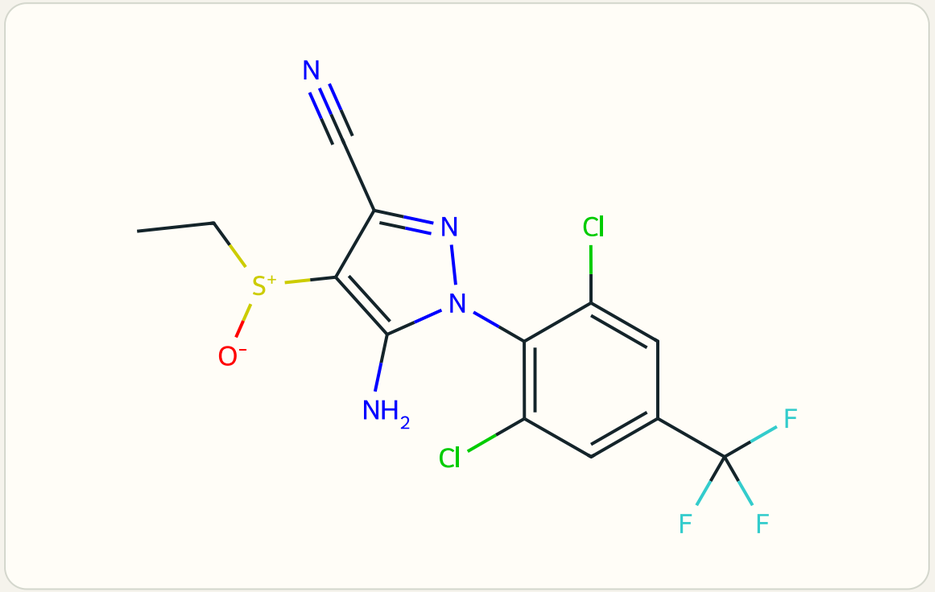
SI Table S26

	RF	LR	Prod	IQP	RBF	MLP
Isotianil	○	○	○	○	○	○
Fluoxapiprolin	○	○	○	○	○	✕
Ethiprole	✕	✕	✕	✕	✕	✕
Cyclobutrifluram	○	○	○	○	○	○
Isoprothiolane	○	○	○	○	○	○
Isocycloseram	✕	✕	✕	✕	✕	○
Spiropidion	○	○	○	○	○	○
Florylpicoxamid	○	○	○	○	○	✕

○ correct decision ✕ error frozen thresholds, no retuning

SI Table S27 · balanced MLP

A nearly identical toxic neighbour was not enough



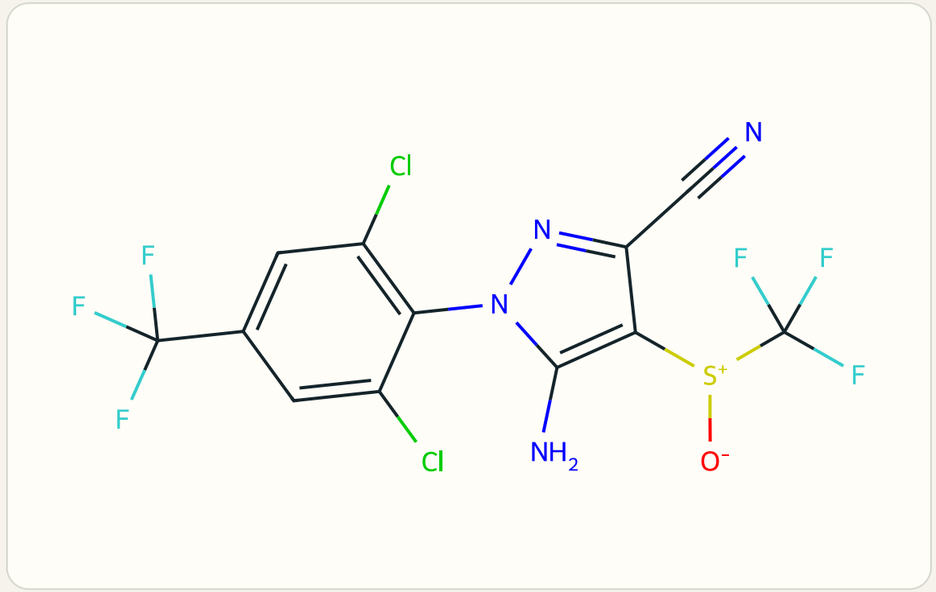
Ethiprole

CR8 · external

● toxic | predicted non-toxic ×6

0.740

Morgan/Tanimoto similarity



Fipronil

Development · ID 387

● toxic | in training

Two findings, and a clear boundary

WHAT WE FOUND

- **No consistent predictive improvement**
  Both fidelity kernels landed very close to the matched RBF.
- **No clearly distinct geometry**
  The Gram matrices turned out highly aligned with each other.
- **Detection remains the challenge**
  Beyond Development, sensitivity falls to 0.082–0.265.

WHAT THIS WORK DOES NOT SHOW

- It does not demonstrate quantum advantage
- It does not evaluate noise or real quantum hardware
- It does not establish clinical or regulatory utility
- It does not generalise beyond X10, these maps and this protocol

Separating kernel changes from representation changes

- **Optimise the encoding scale**

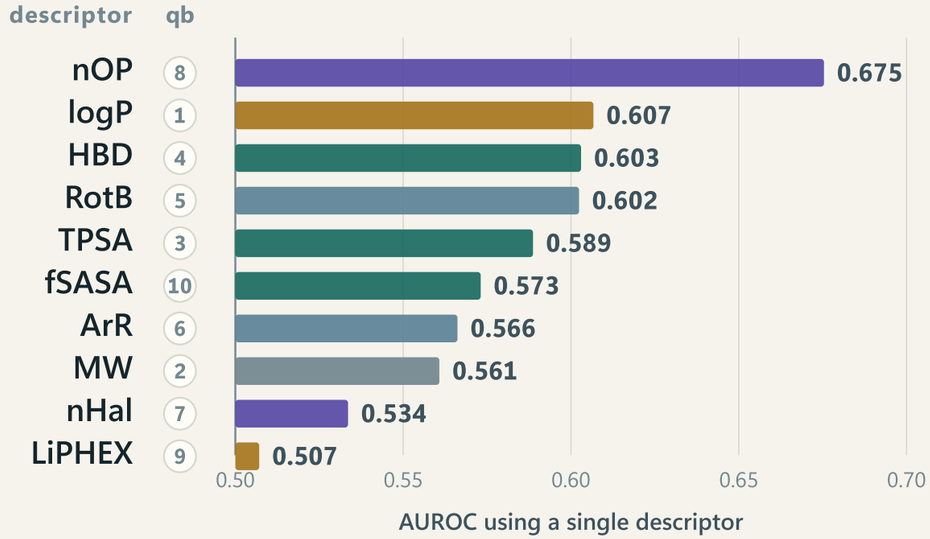
  Bandwidth turned out to be decisive; it deserves to be an object of study rather than a by-product of the search.

- **Train the embedding by kernel-target alignment**

  Instead of fixed maps, learn the map's parameters.

- **Rearrange and widen the qubits**

  Duplicate qubits per descriptor, add more descriptors, or change the order: since the RZZ chain only couples neighbours, which descriptor sits next to which may matter.



SI Table S5 · descriptive

Thank you.
Questions?

DATA AND CODE

Supporting Information archived on Zenodo
doi.org/10.5281/zenodo.23329579

Curated dataset, descriptors and training workflow
github.com/andresquesadag/BeeQ

University of Costa Rica · Costa Rica Institute of Technology